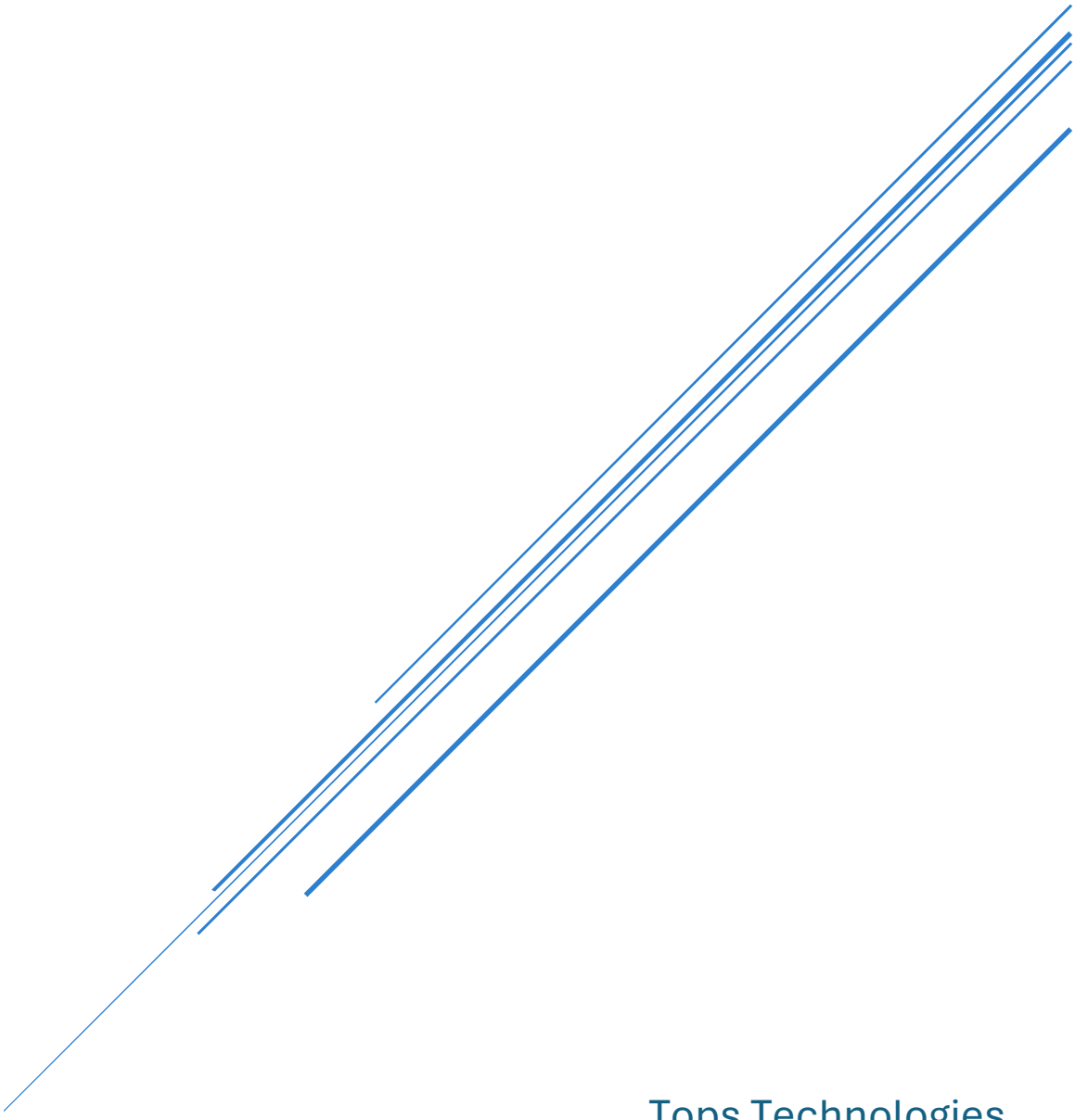
OSI LAYER

Assignment - 3



Tops Technologies
Cyber Security

1. **Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).**

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OSI 7-Layer Model

	Layer & Name	Main Function	Real-World Example
7	**Application Layer** (Provides network services to end users)	Handles user applications and provides the interface for network communication.	**HTTP / HTTPS** (Web browsing)
6	**Presentation Layer** (Formats and translates data)	Ensures data is in a readable format, handles encryption and compression.	**SSL / TLS** (Encryption)
5	**Session Layer** (Manages communication sessions)	Establishes, manages and terminates sessions between applications.	**NetBIOS** (Session management)
4	**Transport Layer** (Ensures reliable data transfer)	Provides end-to-end communication, error recovery, and flow control.	**TCP** (Reliable delivery)
3	**Network Layer** (Handles routing and addressing)	Determines the best path for data and handles logical addressing (IP addresses).	**Router** (Forwards packets)
2	**Data Link Layer** (Handles framing and error detection)	Provides node-to-node communication, handles MAC addressing and error check.	**Switch** (Uses MAC addresses)
1	**Physical Layer** (Transmits raw data bits)	Defines the physical medium, signals, and data transmission rates.	**Ethernet Cable** (Wired transmission)

2. **Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.**
   Hint:Write your answers as 'Scenario: Layer Name (Layer Number)'.

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a) Encrypting data before sending: **Presentation Layer (Layer 6)**

b) Routing packets across networks: **Network Layer (Layer 3)**

c) Converting data to electrical signals: **Physical Layer (Layer 1)**

d) Opening a web page in your browser: **Application Layer (Layer 7)**

e) Detecting and correcting errors in frames: **Data Link Layer (Layer 2)**

3. **Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.**

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App: WhatsApp

When I send a message on WhatsApp, different OSI layers work together to deliver the message to the receiver.

1. **Application Layer (Layer 7**): WhatsApp allows me to type and send messages to another person.
2. **Transport Layer (Layer 4):** Protocols such as TCP help transfer data reliably where TCP is used.
3. **Network Layer (Layer 3):** IP addresses help route data packets from my device across networks to the destination.
4. **Data Link Layer (Layer 2):** Wi-Fi or Ethernet carries frames between devices on the local network.
5. **Physical Layer (Layer 1):** Radio signals or electrical signals carry the data through the network medium.

4. **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

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Application Layer (Layer 7), Transport Layer (Layer 4), or Presentation Layer (Layer 6).

Reasoning:

- **Network Layer (Layer 3):** Since the website's IP address responds to ping, IP connectivity is working at least to that host.
- **Transport Layer (Layer 4):** The website's TCP port (usually 80 for HTTP or 443 for HTTPS) may be blocked or unavailable.
- **Application Layer (Layer 7):** The web server or HTTP/HTTPS service may be down or experiencing problems.
- **Presentation Layer (Layer 6):** An HTTPS/TLS certificate or encryption issue could prevent the browser from connecting securely.